Gen3 EMR Experience Design & EVA Architecture

Companion to the Architecture Handbook V8 — March 2026

Abstract

UX Design Paradigm, Interaction Model, and Engineering Implementation
WebPT AI-Native EMR Platform

Revision History

Date	Version	Author	Description
2026-03-28	V1	Uday Devalla	Gen3 EMR Experience Design and EVA Architecture Initial Release. Companion to the Architecture Handbook V8. Covers the three-mode interaction model, role-specific use cases, mode transition rules, notification design, quick actions and action chips, ambient scribe UX, AI recommendation display and autonomy lane visibility, EVA architecture (conversational interface, not Super Agent), engineering components, intent resolution pipeline, session state, learning loops, and delivery phasing.
2026-04-05	V2	Lal Mohandas	Section 15.1 is expanded with subsections (15.1.1–15.1.3) specifying the Session Service architecture, Context Stack model, panel synchronization, and divergence handling. The original 15.1 paragraph is preserved as the opening. Sections 15.2 (Capability Execution) and 15.3 (Response Generation) are unchanged. Section 17 walkthrough and Section 19 invariants are updated to align with the Context Stack model.

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Overview

This document is organized in two parts. Part I specifies the user experience design paradigm for the Gen3 EMR — the three-mode interaction model, role-specific use cases from the design wireframes, and proposals for mode transition rules, panel coordination, notification tiers, action chips, ambient scribe UX, and how autonomy lane governance appears to users. **The design paradigms and patterns are still evolving, but the core three-mode model will remain intact. As the design paradigms/patterns change, this document will be updated to reflect the decisions made, and their impact on the Part II.**

Part II specifies EVA — the conversational AI interface that powers the left panel — including the critical architectural decision that EVA is a governed conversational interface and not a Super Agent, the six engineering components, the intent resolution pipeline, learning loops, and delivery phasing. However, end users will look at EVA as a Super Agent, and that is fine from a product marketing perspective. We can refer to it as such, but the technical implementation is that of a governed conversational interface.

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PART I

UX Design Paradigm

1. Design Philosophy

The Gen3 EMR is not a chatbot bolted onto a traditional EMR. It is a genuinely integrated paradigm where the AI conversation and the contextual screen are two views of the same underlying operational state. The fundamental tension that every AI-native application faces is that some tasks benefit from conversational interaction, some benefit from spatial/structured interfaces, and many benefit from both simultaneously. The Gen3 design team came up with a brilliant approach that resolves this by letting the user control the balance rather than forcing one mode.

This is an important distinction from both the "chatbot sidebar" pattern (where AI is an afterthought) and the "AI-only" pattern (where structured data is hidden behind conversation). The Gen3 paradigm treats AI conversation and structured context as equal, complementary presentation layers over the same governed platform capabilities.

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2. Three-Mode Interaction Model

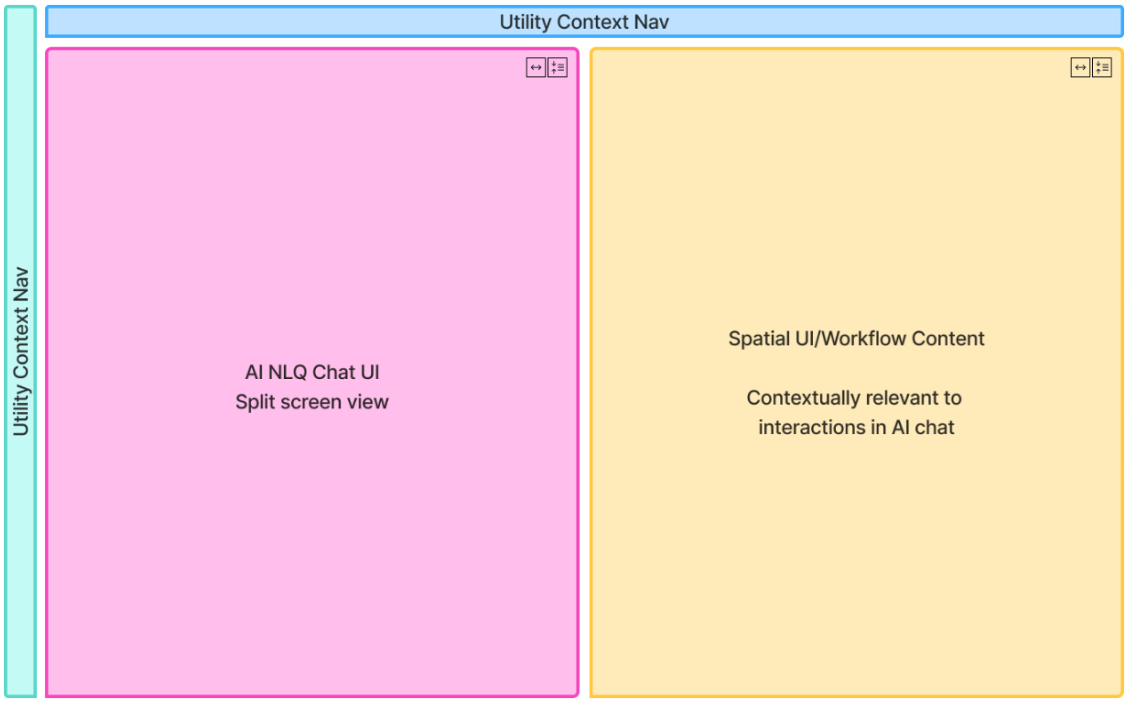
2.1 Hybrid Split-Screen (Default Mode)

Form the March 20th design decision, the default view is a dual-panel layout: AI conversational panel on the left, information-rich contextual screen on the right. The right-side context changes based on the context of the AI/human conversation on the left. This is the primary interaction mode for all roles.

The left panel is where EVA lives — the conversational AI interface. It shows greetings, task summaries, AI-generated insights, inline data tables, action chips for follow-up, and a text input at the bottom with quick action suggestions. The left panel also contains a left sidebar with vertical icon navigation for system-level functions.

The right panel is the structured, information-rich view. Depending on context, it shows patient dashboards, daily calendars, referral management workflows, analytics dashboards, SOAP notes, or any other data-intensive screen. A content-based contextual navigation bar (the "green bar") appears at the top of the right panel, providing traditional menu navigation that reflects the content currently displayed.

Between the two panels, expand/collapse controls allow the user to resize or collapse either side.



2.2 Collapsed AI Mode

The user can collapse the left AI panel to a small chat input bar at the bottom of the screen. The contextual screen expands to full width. The content-based contextual navigation bar (green bar) appears at the top, providing traditional tab-based navigation reflecting the current content: Patients, Today's Schedule, Search Patients, Documentation, Assigned Cases (for providers); Reports, Staffing, RCM Reports, Discharge Forecast, Workload Forecast, Risk & Adherence, Documentation Efficiency, Claims & AR (for owners/analysts).

This mode serves users who want to focus on structured data and workflows, with AI available on demand. The collapsed chat widget remains visible at the bottom, so the user can ask a question or

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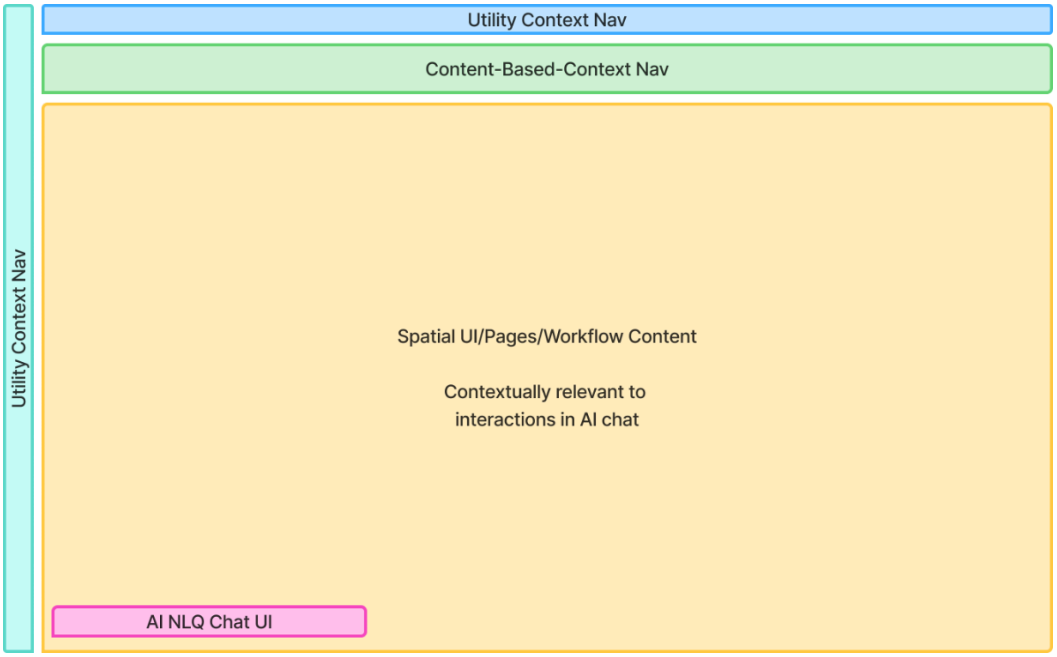
see notifications at any time. When the user has collapsed the AI panel, an AI Recommendation card can still appear within the right panel's content (as shown in the analytics view), keeping AI intelligence present without requiring the conversational panel.



2.3 Collapsed Context Mode

The user can collapse the right contextual panel to a thin strip. The AI conversational panel expands to full width. This mode is for users who want an AI-first interaction without the contextual screen taking space — useful for extended conversational sessions, complex multi-turn questions, or when the user is working on a device with limited screen real estate.

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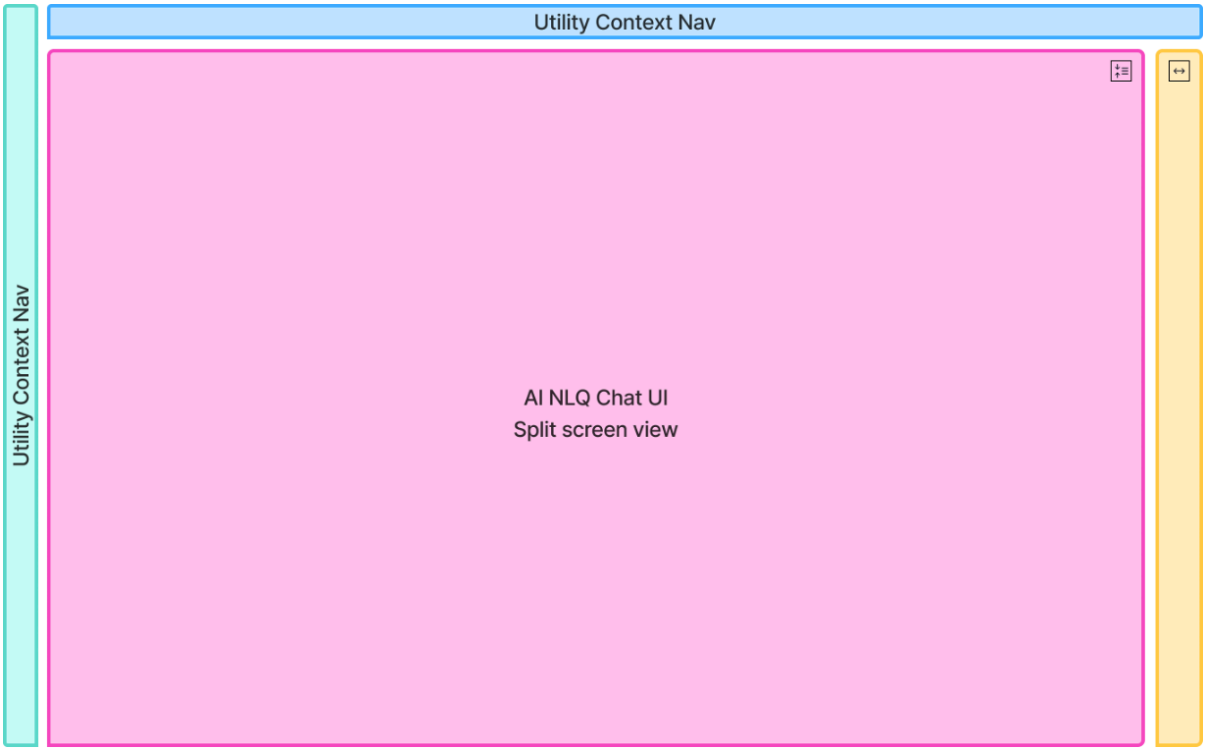

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3. Role-Specific Use Cases

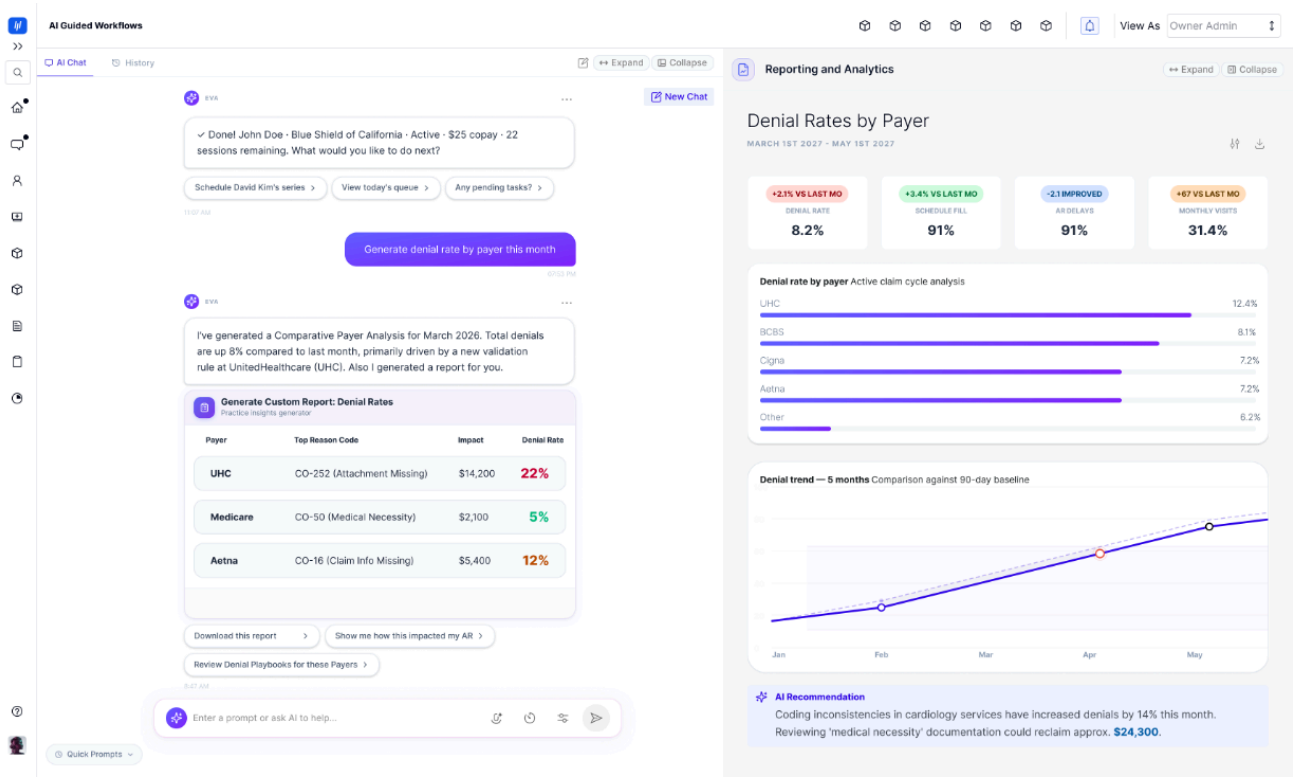
The original design wireframes (from 3/20 discussion) demonstrate how the three-mode paradigm adapts to four distinct personas: clinic owner, front office staff, provider (therapist), and referral intake coordinator. Each role uses the same foundational interaction model but with different default content, priorities, and AI behaviors.

3.1 Clinic Owner — Analytics

Hybrid Mode

The left panel shows EVA's analysis of denial trends, presented as a conversational response with a structured inline table showing payer, top reason code, financial impact, and denial rate. Below the response, action chips offer follow-up options: "Download this report" / "Show me how this impacted my AR" / "Review Denial Playbooks for these Payers." At the bottom, quick actions suggest common owner queries.

The right panel displays a full analytics dashboard: KPI cards at the top (Denial Rate, Collections, Clean Claim Rate, Avg Days to Payment), a denial rate by payer bar chart, a 5-month denial trend line chart, and an AI Recommendation card at the bottom. The AI Recommendation card states a specific finding ("Coding inconsistencies have increased denials by 14%") with a quantified action opportunity ("could reclaim approx. $24,300"). This is the conversational analytics model from C.15 rendered in the dual-panel paradigm: the AI provides summary and follow-up in conversation, while the dashboard provides the full governed metric visualization.



Collapsed AI Mode

The content-based contextual nav bar shows tabs: Reports, Staffing, RCM Reports, Discharge Forecast, Workload Forecast, Risk & Adherence, Documentation Efficiency, Claims & AR. A left sidebar shows an AI-Generated Reports list with report titles that can be selected. The main area

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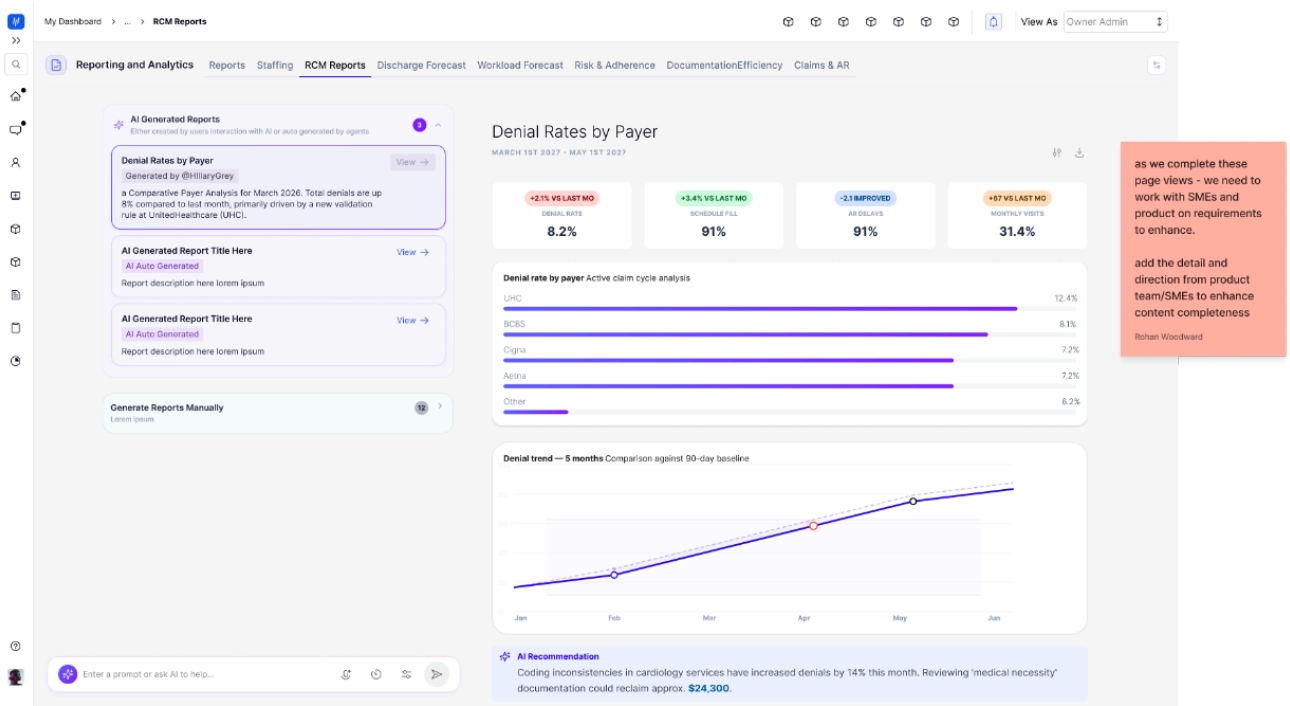
shows the full dashboard. The collapsed chat input bar sits at the bottom. The AI Recommendation card remains embedded in the dashboard content. A "Generate Reports Manually" option is available in the sidebar for users who prefer traditional report creation.



3.2 Front Office — Daily Calendar

Hybrid Mode

The left panel shows EVA's morning greeting: "Good Morning, Sarah. You have 3 priority tasks today" with expandable task cards (a new referral intake, a pending eligibility check, and a series scheduling action). Quick actions at the bottom suggest common front office actions: "John Doe is here" / "Denial rates by payer."

The right panel shows a multi-provider daily calendar with time slots, patient names, and status indicators. Status badges appear inline on each appointment: "Validated" (green), "Co-Pay Due" (yellow), "Insurance Pre-Auth Fail" (red), "Overdue Balance" (orange). These status indicators are the visible output of Green-lane agents that ran eligibility checks, balance lookups, and authorization verifications in the background before the staff member's shift began. The next patient arriving (e.g., Alicia Roman at 8:45 AM) is highlighted with additional context.

The calendar view bridges AI intelligence and traditional EMR structure: the calendar is a familiar artifact that front office staff know how to read, but it's enriched with agent-computed intelligence (status badges) that would not exist in a traditional system.

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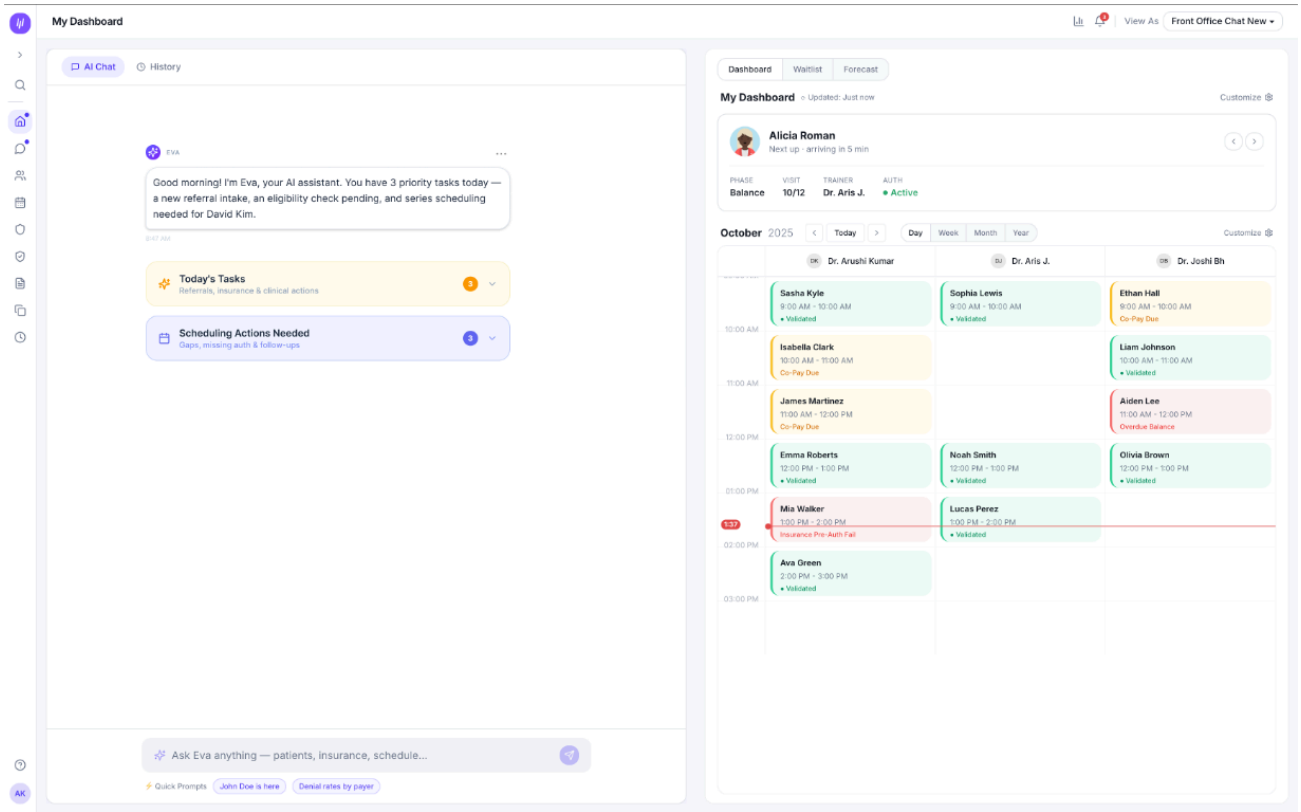

3.3 Provider — Patient Session

Hybrid Mode

The left panel shows EVA's session preparation: "Good Morning, Dr. Aris. You have 6 patients today, and I've prepared your schedule. 2 post-op cases may need closer review." Below this, a "Documentation Update — Completed" card summarizes the most recent progress: "Elena Richardson has shown progression in her recovery milestones. Over the past 3 visits, her pain levels have decreased from 7/10 to 4/10." This is ambient scribe output presented as a reviewable summary rather than raw dictation.

The right panel shows a Pre-Visit Patient Summary for the next patient: patient name, condition, visit count, a body diagram highlighting affected areas, pain assessment scores, current goals, and care plan summary. Action items appear as cards: "Reassess Functional Mobility" and "Update Home Exercise Program." This pre-visit summary is built from the patient's episode context in the Care-to-Cash graph, assembled by the Context Assembly Service and presented through the BFF.

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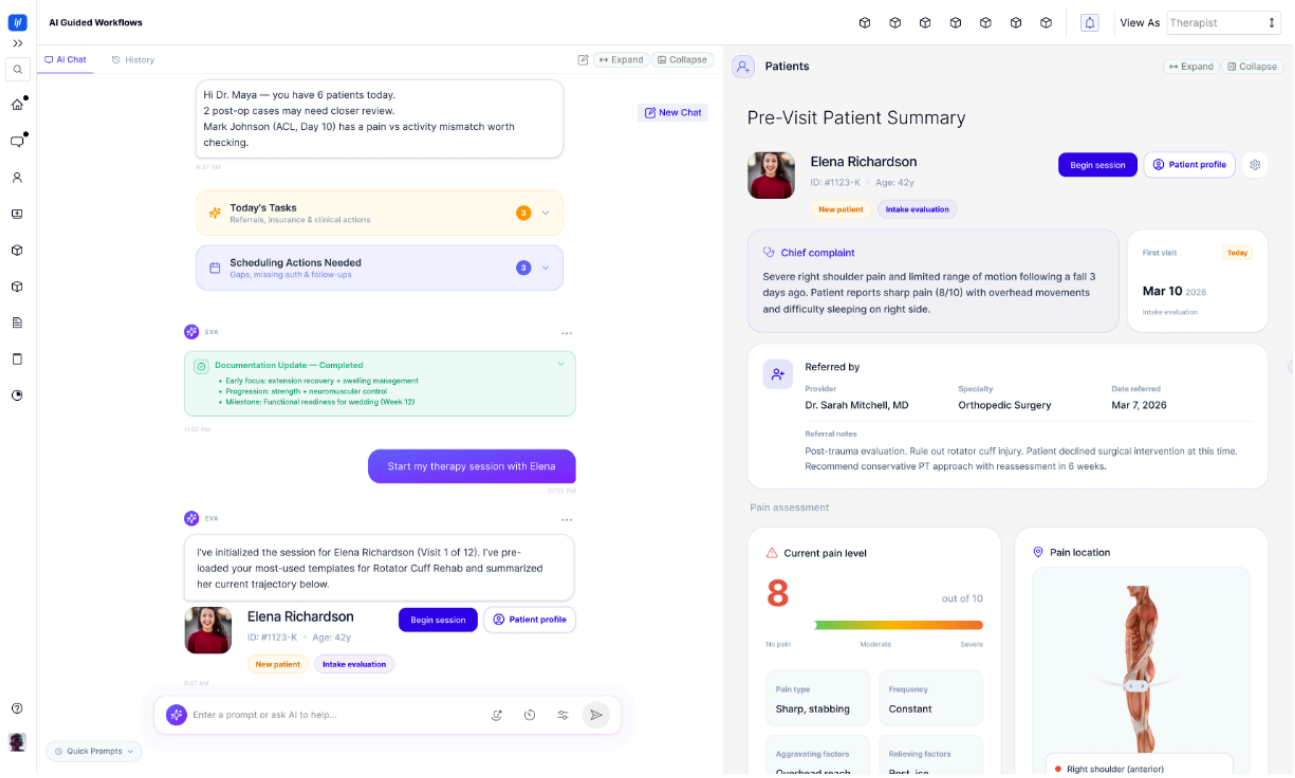

Collapsed AI Mode

The content-based contextual nav bar shows tabs: Patients, Today's Schedule, Search Patients, Documentation, Assigned Cases. The left portion shows Today's Patients with status counters (8 Today, 2 In Clinic, 5 Waiting, 1 Urgent) and a filterable patient list with session times, status badges (Urgent, Checked In), and timing overlap indicators. The right portion shows the same Pre-Visit Patient Summary. The minimized chat bar sits at the bottom. This is a traditional EMR layout enriched with AI-computed context — users who prefer structured navigation can work here while AI remains available but unobtrusive.

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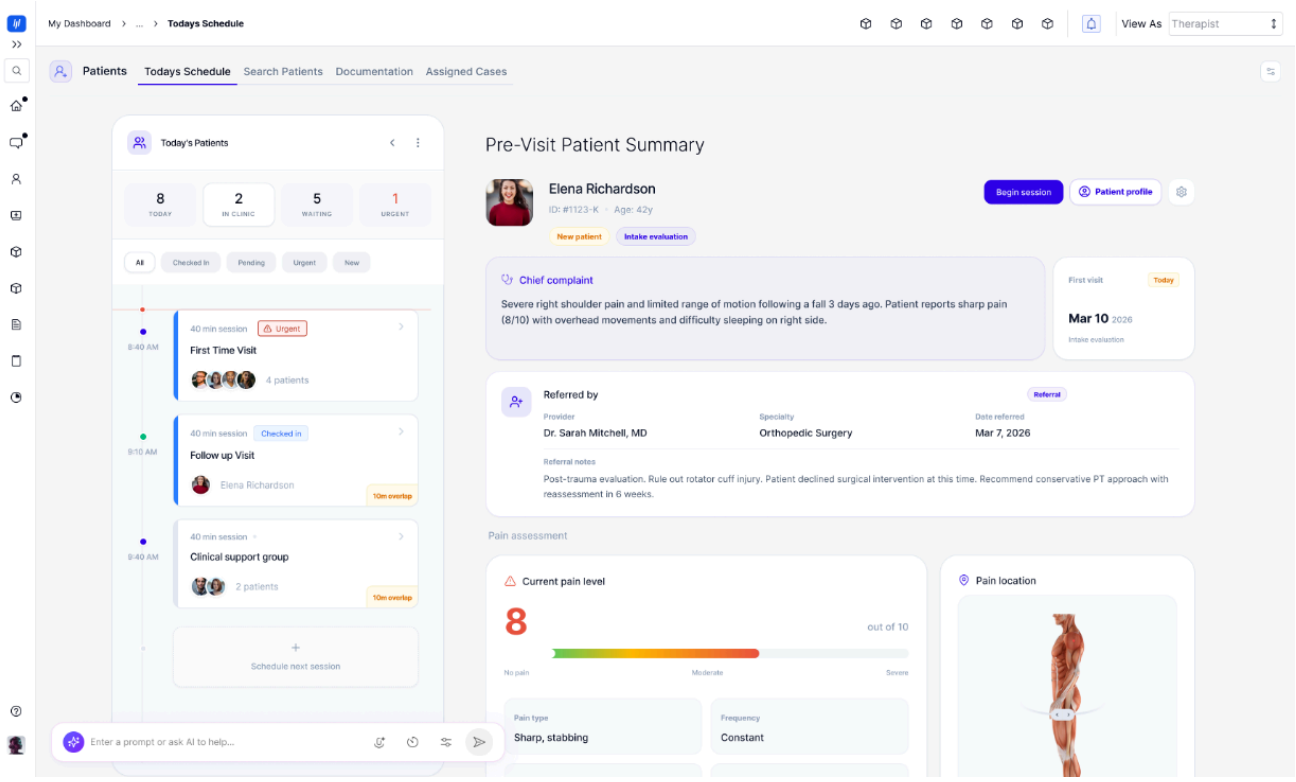

3.4 Front Office — Referral Intake

Hybrid Mode (Standout Use Case)

This is the most compelling wireframe and the one that most fully demonstrates the V8 architecture working end-to-end.

The left panel shows a conversational referral intake flow: EVA has received and processed a referral. The user says "John Doe is here" and EVA responds with a patient match, referral context, and a compact patient card showing status. A "Retrieve in full view" link offers to open the referral workflow in the right panel. Suggested follow-up questions appear: "Is this John Doe's first visit?" / "Check John Doe's insurance" / "Any notes on the referral?"

The right panel shows a Referral Management workflow screen that is the architectural highlight of the design. At the top, a patient card shows John Doe with referring physician, phase (Not Yet Started), visit count (0/12), and authorization status (Pending). Below this, completed agent actions appear in green with checkmarks: "Identity Verified" and "Insurance Confirmed," each with AI annotation bubbles explaining what was pre-filled and what the user should validate (e.g., "Copay analysis: $30 copay per visit based on plan type"). These are Green-lane agent outputs that were auto-executed and are now presented for human validation.

An incomplete item appears in orange: "Clinical Triage Incomplete" with pre-filled referral information from the referring physician. This requires human input — either because the information is missing or because clinical triage is a Yellow/Red-lane action requiring human judgment.

At the bottom, a Dynamic Scheduling Hub offers three AI-computed scheduling options: "Soonest" (earliest available slot), "Therapist Match" (best provider match based on specialty and patient needs), and "Clinic Optimal" (best slot considering overall clinic schedule). These are agent-generated recommendations with visible choice architecture rather than a single forced suggestion.

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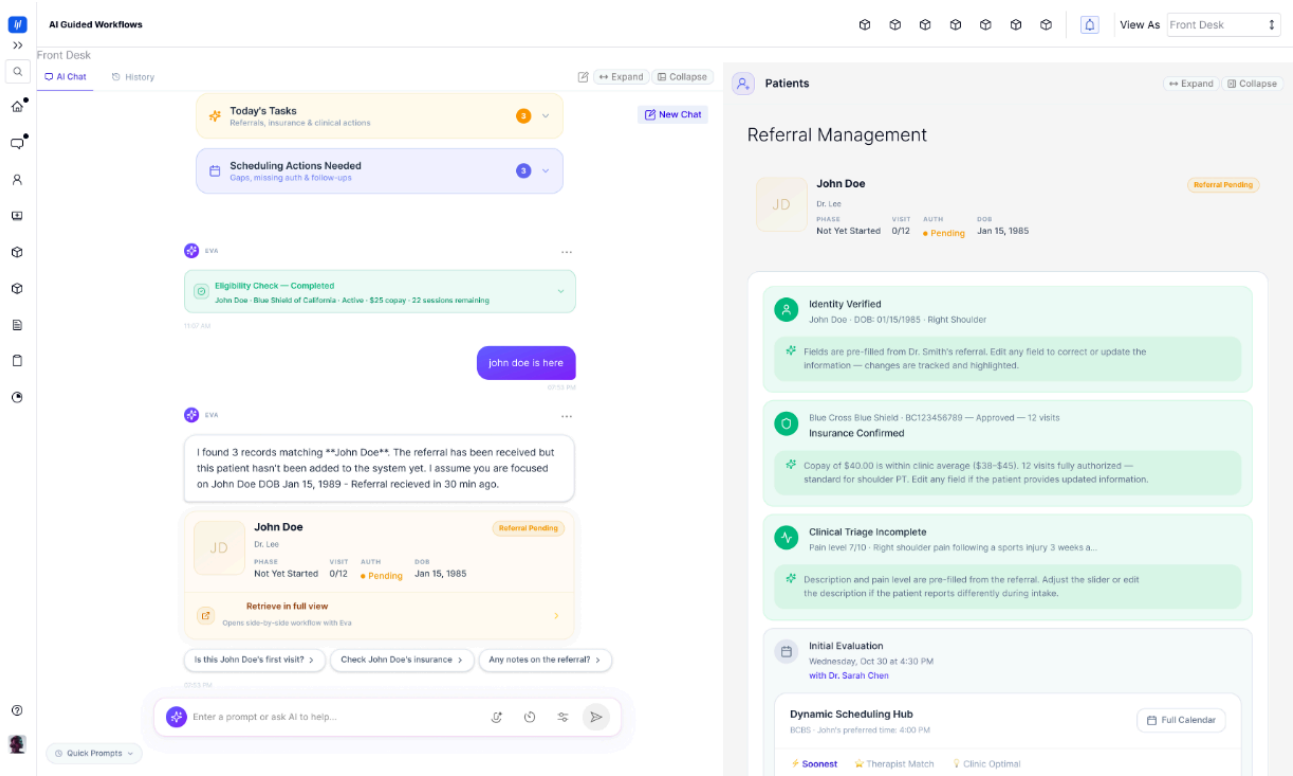
This screen demonstrates the full autonomy lane model at the UX layer: Green-lane actions (identity verification, insurance confirmation) were completed automatically and shown as done. Yellow/Red-lane actions (clinical triage) are presented as requiring human input. Scheduling recommendations are presented as options, not decisions. The user sees what the AI did, what it recommends, and what requires human judgment — without needing to understand anything about lanes, agents, or context packs.



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4. Mode Transition Rules - Proposal

The following proposed rules govern how the three modes interact. These should be finalized during the design review and reflect confirmed product decisions.

4.1 Right Panel Follows Conversation

When the user is in hybrid mode and the AI conversation shifts to a different topic or entity, the right panel navigates to the new context automatically. The right panel is contextual to the AI conversation. For example, if the user is viewing a patient's referral and asks EVA about denial rates, the right panel transitions to the analytics dashboard. If they then ask about a specific patient, the right panel navigates to that patient's record.

4.2 Collapsed AI Expands on Interaction

If the user is in collapsed-AI mode and types a question, or if the AI needs to produce an action, task, or result that requires the conversational panel, the left panel opens and the layout returns to hybrid mode. The system does not force the user to manually expand the AI panel before asking a question. We need to ensure that this auto transition is smooth and easy.

4.3 Proactive Expansion: Notification Badge, Not Panel Change

When the AI panel is collapsed and an agent completes a background task, EVA does not proactively expand the panel. Instead, she shows a notification badge on the collapsed chat widget. If needed the chat widget can expand a bit to show the notification badge (perhaps double the default size)The user can open it when they are ready. This is a critical healthcare UX decision: during clinical work (provider documenting a SOAP note, for example), unexpected layout shifts are disorienting and erode trust. A panel that forces itself open mid-documentation interrupts the therapist's workflow and creates frustration.

The only exception that would warrant a forced interruption is a genuinely urgent, safety-relevant alert (such as a drug interaction warning during a clinical session). Even then, the preferred pattern is a modal overlay that demands attention rather than a panel expansion, because a modal is an explicit interruption the user must acknowledge, while a panel expansion is a subtle layout change they might not notice. While the use cases will be very few, We need to debate this and finalize the approach for such urgent alerts.

4.4 Unsaved Work Protection

If the user's follow-up question in the AI conversation would navigate the right panel away from a screen with unsaved work, EVA prompts the user to save or discard before proceeding. The prompt appears in the AI conversation panel: "You have unsaved changes in the referral form. Would you like to save before I switch to the analytics view?" This prevents accidental data loss while keeping mode transitions fluid.

4.5 User Can Always Override

The user can manually expand or collapse either panel at any time, regardless of what the AI is doing or what context is active. Mode control is fundamentally user-owned. The system suggests transitions; the user controls them.

4.6 User Acts Independently in the Context Panel

When the user completes an action directly in the right panel without going through EVA — for example, manually updating a field, closing a work item, or navigating to a different screen using the

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contextual nav bar — EVA must acknowledge the new state. The EVA Session Service detects right-panel state changes (via events emitted by the MFE) and updates the focus entity and conversation context accordingly. If the user closes a work item in the right panel, EVA's next response should reflect that the item is closed rather than suggesting actions on it.

The key principle: EVA and the context panel share a synchronized view of the current state. Neither panel should show stale information when the other has already moved forward. The BFF's session state is the single coordination point.

4.7 Does Panel Interaction Vary by Workflow or Persona?

The foundational panel interaction rules (right follows conversation, unsaved work protection, notification badges) are consistent across all workflows and personas. However, the default panel emphasis and content density will vary by role:

Providers during patient sessions: Right panel emphasis — the SOAP note and patient record are the primary workspace. EVA is a preparation and assistance layer. The ambient scribe runs in the background — capturing audio, processing segments, extracting structured data — but its output appears as inline assists within the clinical screen itself, not as messages in the left conversation panel. Section 7.1 goes into more detail on the Ambient Scribe interactions.

Front office during intake: Balanced hybrid — EVA guides the workflow conversationally while the right panel shows the structured referral management screen. Both panels are active and equally important.

Owners reviewing analytics: Left panel emphasis — the owner asks questions conversationally and EVA provides analysis. The right panel supports with dashboards but the conversation drives the flow.

Billers working queues: Right panel emphasis — the work queue and claim details are the primary workspace. We will need to work with Candid to see how EVA can assist with context and recommendations, but the biller works primarily in structured views, which are part of Candid.

These variations affect default mode and content, not the interaction rules themselves. A provider in collapsed-AI mode follows the same transition rules as a front office user in hybrid mode.

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5. Notification Design Proposal

5.1 Why Not User-Configurable

Users should not configure their own notification preferences. This is a deliberate product decision. User-configurable notification settings fail predictably: users either select all notifications (defeating the purpose of tiers), select none (missing critical alerts), or configure once and never update as new notification types are added. In healthcare, notification fatigue is a documented patient safety concern — Epic and Cerner have extensive literature on alert fatigue contributing to clinicians ignoring genuinely critical warnings.

The alternative: the platform governs notification tiers based on the user's role, their current activity context, and the urgency of the event. An admin or operations lead can tune tier rules at the practice level if specific events are too noisy, but individual users do not manage their own notification settings.

5.2 Three Notification Tiers

Tier	Behavior	UX Signal	Examples
Tier 1: Ambient	No interruption. Badge count only.	Notification bell gets a count. Collapsed chat widget gets a subtle dot. User sees it when they naturally look.	Eligibility verified, batch claims submitted, daily report generated, Green-lane agent completed background task.
Tier 2: Nudge	Gentle attention. Brief toast + badge.	Brief toast notification (disappears after seconds) plus a more prominent badge on the chat widget. User notices without being forced to act.	SLA approaching on work item, AI recommendation ready for review, patient checked in, Yellow-lane output awaiting review.
Tier 3: Interrupt	Demands attention. Persistent banner.	Persistent notification bar across top of screen or modal overlay. User must acknowledge. Rare by design.	Authorization about to expire for patient currently in clinic, critical claim rejection needing same-day resubmission, clinical safety concern.

5.3 Tier Assignment Logic

The tier is determined from the event's metadata and the user's role/context, not from user preferences. The workflow engine and agent contracts already know whether something is urgent (SLA breach imminent), important (Yellow-lane review needed), or informational (Green-lane action completed). The Notification Router in the BFF maps this to the appropriate tier. If everything is Tier 3, nothing is Tier 3 — the system must be designed so that interrupts are genuinely rare, preserving their urgency signal.

5.4 Notification Rendering by Mode

Hybrid mode: Tier 1 notifications appear as badges on the notification bell. Tier 2 notifications appear as a toast plus a new message in the AI conversation: "Just completed: Elena Richardson's eligibility has been verified. 12 visits authorized." Tier 3 notifications appear as a persistent banner or modal.

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Collapsed AI mode: Tier 1 shows a subtle dot on the collapsed chat widget. Tier 2 shows a more prominent badge plus a brief toast. Tier 3 shows a persistent banner across the top of the full-width screen or a modal. The AI panel does not force-expand for Tier 1 or Tier 2.

Collapsed Context mode: All tiers render in the AI conversation panel since it is already expanded. Tier 3 may additionally show a banner or modal.

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6. Quick Actions and Action Chips Proposal

6.1 Quick Actions

Quick actions (referred to as "quick prompts" in early wireframes) appear at the bottom of the AI conversation panel as suggested starting actions. The term "Quick Actions" is preferred because it communicates that these are things the user can do, not prompts they are writing. They are not hardcoded UI elements. They are governed outputs generated from the user's role, current context, and practice patterns.

On day one, quick actions are generic role-based defaults: for front office staff, "John Doe is here" / "Check eligibility" / "Pending referrals." By week four, the actions reflect the user's actual workflow patterns learned from the Layer 1 user preference profile: if Sarah always starts Monday mornings by checking referrals, Monday's quick actions lead with referrals. By month three, practice-level intelligence (Layer 2) further refines the actions based on the practice's current operational state: if there's a spike in Aetna denials this week, the actions might include "Review Aetna denial trends."

6.2 Action Chips

Action chips appear below every EVA response as suggested follow-up actions: "Download this report" / "Show me how this impacted my AR" / "Review Denial Playbooks for these Payers." These are generated from the context pack's allowedActions field and the agent's suggested follow-up actions. The architecture treats action chips as part of the agent's output schema — the context pack shape includes suggestedFollowUpActions and the BFF renders them as clickable chips. This keeps suggestions relevant, governed, and traceable.

Action chips serve discoverability: they show users what they can do next without requiring them to know the platform's full capability set. A new front office hire who has never used the system can follow the suggested actions and accomplish their work on day one.

6.3 "Retrieve in Full View" Links

When EVA shows an entity or workflow result in the conversational panel, a "Retrieve in full view" link offers to open the corresponding structured view in the right panel. This is the bridge between conversational and structured interaction: the user gets a quick answer in conversation, and if they need the full detail, one click opens the complete screen. The link triggers a right panel navigation command through the Panel Coordinator.

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7. Ambient Scribe UX Proposal

The provider session wireframes demonstrate how the ambient scribe specification (Handbook Section 9.10.3) manifests in the user experience.

7.1 Three-Phase Ambient Scribe Experience

The ambient scribe experience follows a three-phase model aligned with the provider's natural workflow. The key principle: during an active clinical session, the therapist's attention belongs on the patient and the clinical record, not on a conversation panel.

Pre-session (hybrid mode). Before the encounter begins, EVA prepares the workspace in the left panel: session prep summary, patient context, documentation updates from prior visits, and any flags requiring attention ("2 post-op cases may need closer review"). The right panel shows the Pre-Visit Patient Summary with pain assessment, body diagram, care plan, and goals. Both panels are active. The provider reviews, absorbs the context, and is ready to begin.

During session (collapsed AI). When the provider enters the active encounter, the AI panel collapses and the right panel expands to full width, showing the SOAP note, patient record, ROM measurement tools, and exercise documentation. The ambient scribe runs in the background — capturing audio, processing segments, extracting structured data — but its output appears as inline assists within the clinical screen itself, not as messages in the left conversation panel. Documentation completeness nudges appear as subtle contextual indicators embedded in the note structure: a gentle highlight on the Assessment section indicating "functional limitation language not yet documented," a small badge on the Plan section suggesting "treatment frequency rationale may strengthen this note," or auto-populated ROM values extracted from the session that the clinician can accept or edit in place. These assists are contextual to where the clinician is working. They do not require the clinician to shift attention to a separate panel, and they do not force the AI panel open.

Post-session (hybrid mode returns). When the therapist ends the encounter, the AI panel reopens with EVA's documentation summary: progression milestones, pain level changes, goal progress, documentation completeness score, and any items that would strengthen the note before sign-off. This is where the Documentation Update card belongs — after the session as a reviewable summary, not during the session as streaming output. The provider reviews, accepts or edits, and signs the note. Canonical events (ClinicalNoteSigned, EvidenceSpanLinked) fire only after sign-off.

7.2 Clinician Control Boundary

The critical UX rule: ambient scribe output is always presented as a draft for clinician review. The therapist sees, edits, accepts, or ignores suggestions. The final note remains under clinician control and crosses a normal sign-off boundary. Durable consequences (canonical events like ClinicalNoteSigned, downstream billing readiness) begin only after clinician acceptance. The UX makes this boundary visible: draft content is visually distinct from finalized documentation, with clear accept/edit controls.

7.3 PT-Specific Documentation Patterns

Because this is a rehab EMR, the ambient scribe’s UX is optimized for therapy documentation patterns: SOAP structure (Subjective/Objective/Assessment/Plan), functional limitation language, measurable goals and progress toward goals, intervention and exercise capture, treatment frequency/cadence rationale, and discharge readiness cues. The documentation completeness signals in the AI panel are specific to PT billing requirements, not generic clinical documentation.

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8. AI Recommendation Display and Autonomy Lane Visibility

8.1 How Autonomy Lanes Appear to Users

Users never see the terms "Green lane," "Yellow lane," or "Red lane." But the governance model is visible in the UX through three distinct interaction patterns:

Completed actions (Green lane): Shown as done items with green checkmarks and an AI annotation explaining what was pre-filled. The user can review but does not need to approve. Examples: "Identity Verified ✓" with a note explaining the matching logic; "Insurance Confirmed ✓" with copay analysis. These appear in the referral intake right panel.

Recommendations requiring review (Yellow lane): Shown as suggested actions that the user can accept, reject, or modify. Examples: appeal draft with an edit button, schedule repair proposals with multiple options (Soonest / Therapist Match / Clinic Optimal), documentation nudges from the ambient scribe. The user sees the recommendation and the reasoning; they choose the action.

Human-required decisions (Red lane): Shown as action items that only a human can complete. Examples: clinical triage for a new referral (shown as "Clinical Triage Incomplete" in orange), note sign-off, final claim submission approval. The system does not attempt these — it presents the required information and waits for human action.

8.2 AI Recommendation Cards

AI Recommendation cards appear in both panels. In the AI conversation panel, they appear as inline messages with action chips. In the right contextual panel, they appear as embedded cards within dashboards or workflow screens (e.g., the "Coding inconsistencies have increased denials by 14%" card in the owner analytics view). Both renderings include: the recommendation itself, the supporting evidence or metric, a quantified impact where applicable, and one or more action options.

8.3 Human Override Recording

When a user rejects or modifies an AI recommendation, the action emits a HumanOverrideRecorded event that captures: what was recommended, what the user chose instead, and the rationale if provided. This feeds into the decision trace model (Handbook Section 9.13) and the Layer 3 learning loop (outcome-linked learning), enabling the platform to improve recommendations over time based on human judgment patterns.

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9. EVA Processing State Display Proposal

When EVA is working on a request that takes more than a moment — invoking a workflow, assembling context, waiting for an agent result — the user needs to know that something is happening. The design decision is to keep this simple.

9.1 Default: Contextual Status Indicator

EVA shows a brief, contextual status message with an animated indicator: "Checking eligibility..." or "Looking up denial details..." or "Preparing your schedule summary..." These messages are generated from the capability being invoked, not from the internal pipeline stages. The user sees what EVA is doing in business terms, not what the system is doing technically.

The user does not see the internal reasoning pipeline (input processing, context gathering, planning, tool selection, execution, synthesis, response generation). That level of detail belongs in the observability layer for engineers, not in the user-facing experience. Showing pipeline stages adds visual complexity without adding user value — a front desk user checking in a patient does not benefit from knowing that the Context Assembly Service is performing a graph traversal.

9.2 Exception Handling

When EVA cannot complete a task, she says so clearly with a next step: "I wasn’t able to verify eligibility for this patient — the payer system isn’t responding. You can try again in a few minutes, or verify manually through the payer portal." The exception message includes: what failed in plain language, why (if known), and what the user can do instead. EVA does not show error codes, stack traces, or technical failure details in the conversation panel.

9.3 Long-Running Tasks

For tasks that take more than a few seconds (e.g., batch operations, complex analytical queries, appeal draft generation), EVA acknowledges the request immediately and provides a progress update: "I’m working on the denial analysis — this usually takes about 30 seconds." When the result is ready, it appears in the conversation thread. If the user has navigated away or collapsed the AI panel, the result arrives as a Tier 1 or Tier 2 notification depending on urgency.

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10. Open Structural Design Decisions – Pending Design Review

The following structural patterns were identified during design review and require resolution before Phase 0 implementation. They are documented here as open decisions with options and trade-offs so the product and design team can finalize them.

10.1 Where Do Next Steps Live?

When a user completes a task (e.g., finishes a referral intake), where should the suggested next steps appear? This pattern affects the Response Generator's output schema, the Panel Coordinator's behavior, and user expectations about where to look for guidance.

Option A — Next steps always come from EVA (left panel): After the user completes the intake in the right panel, EVA says "Intake complete. The consent form is awaiting the patient's signature — would you like me to send a reminder? Your next task is series scheduling for Maria Lopez." All workflow guidance flows through EVA. This is consistent with EVA's role as the conversational workflow guide and gives the user a single place to look for what to do next.

Option B — Contextual next steps in right panel, external next steps from EVA: If the next step is contextual to the current task (e.g., "send consent form reminder" after intake), it appears in the right panel as an action button within the workflow screen. If the next step is a different task entirely (e.g., "move on to Maria Lopez's scheduling"), it comes from EVA in the left panel. This keeps task-contextual actions co-located with the task they relate to, while EVA handles cross-task transitions.

Trade-offs: Option A is simpler and creates a single pattern users can rely on. Option B reduces left-panel clutter and keeps related actions together, but risks the user looking in two places for next steps. If Option B is chosen, the rule must be clearly defined: contextual follow-up actions (same subject, same workflow) appear in the right panel; cross-subject or cross-workflow transitions come from EVA.

10.2 Task Organization and State Clustering

When EVA presents a list of tasks in the left panel (e.g., "Scheduling actions needed"), how should completed tasks behave? Two options:

Status-based regrouping: Completed tasks move to a "Completed" section. "Scheduling actions needed" shows only incomplete items; a separate "Completed scheduling" section appears below. This keeps the active task list clean but means items move between sections as their status changes, which can be disorienting.

Category-based with inline status: The section is called "Scheduling" and contains all scheduling-related tasks regardless of status. Each task shows its own status badge (completed, in progress, waiting, incomplete). This keeps related tasks together and avoids items jumping between sections.

The category-based approach is recommended because it reduces cognitive load — the user always knows where to find a scheduling task. But this is a UX validation question that should be tested with users.

10.3 Today's Tasks Drawer Placement

The current wireframes show today's tasks as cards within the EVA conversation flow. An alternative proposed during design review is to move the Today's Tasks to a persistent drawer at the top of the left panel (where History currently sits), making it an always-accessible control rather than

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something embedded in the conversation thread. This would make the task list a primary navigation element that the user can reference at any time without scrolling through conversation history.

If adopted, the History function needs a new home — potentially as a menu item in application preferences or as a conversation search function accessible from the input bar. The value of persistent History as currently implemented should be evaluated: is conversation history something users actually reference, or is it an artifact of chat-based UX conventions?

10.4 Action Button Labels

Feedback identified that generic labels like "Start Task" are less useful than specific action labels like "Schedule Patient" or "Verify Eligibility." Where screen real estate permits, action buttons should use the specific action name. Where space is constrained (e.g., task list cards), a compact icon (play button or arrow) with tooltip is acceptable. The Response Generator should output specific action labels as part of the suggestedFollowUpActions schema, not generic verbs.

10.5 EVA capabilities in the back-office workflows

As the structured views in the right contextual panel will be Candid screens, how much can EVA drive the workflow and user interaction – we will need to work with Candid to decide that capability, as that capability is primarily provided as work queues. Will there is ability to EVE to understand the work queue and provide a conversational interface to the back-office users? Do we want that?

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11. Architectural Impact of the UX Paradigm

The design paradigm is architecturally compatible with V8 as-is, with one extension and several validations.

11.1 New Requirement: Richer BFF Session Model

The hybrid split-screen model means the BFF is simultaneously serving the AI conversation panel (streaming responses, action chips, inline data cards) and the contextual panel (structured data views, dashboards, forms) from a unified session state. The BFF needs to maintain an Experience Session that tracks: what the AI is discussing, what the right panel is showing, what mode the user is in, what entity is in focus, and whether there is unsaved work in the right panel. This is a richer BFF contract than Appendix C.7 currently specifies. The EVA Session Service (described in Part II) provides this.

11.2 Validations

Ambient scribe (Section 9.10.3): The provider session views show exactly the behavior that the handbook specifies. No architectural change needed.

Conversational analytics (C.15): The owner analytics views demonstrate the metric-governed mode and investigative drill-through path. The semantic layer serves both the AI conversation and the dashboard from the same governed metrics. No architectural change needed.

Green-lane agent model (Sections 9.5–9.9): The referral intake view demonstrates Green-lane agent outputs presented for human validation. No architectural change needed.

Human-in-the-loop (Section 10.12): The referral intake view shows Yellow/Red-lane items requiring human input alongside Green-lane completed items. No architectural change needed.

Context Assembly Service (Section 8): The pre-visit patient summary on the provider screen is a context pack rendered through the BFF. No architectural change needed.

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PART II

EVA — Architecture & Engineering Implementation

12. What EVA Is and What EVA Is Not

12.1 The Critical Design Decision

EVA is the conversational interface to the platform's governed capabilities. She is not a Super Agent that orchestrates work autonomously.

From the user's perspective, EVA feels like a super agent. She understands requests, coordinates responses, shows results, and suggests next actions. The user experience is indistinguishable from interacting with an intelligent orchestrator.

From the engineering perspective, EVA is a thin conversational layer in the BFF that: parses user intent, resolves entities against platform state, invokes the appropriate workflow or API, streams agent outputs back as they complete, renders results in both panels, and generates contextual follow-up suggestions.

12.2 Why Not a Super Agent

If EVA were a literal Super Agent receiving requests and deciding which sub-agents to invoke, the workflow logic would live inside EVA's prompt reasoning rather than in the durable, auditable, replayable workflow engine. This creates four critical problems for the platform:

Durability: If EVA's reasoning session crashes, the state is lost. The workflow engine (Temporal) is durable by design — workflow state survives crashes, restarts, and redeployments.

Auditability: A workflow definition is a governed, versioned artifact that can be reviewed, tested, and audited. Prompt-based orchestration produces different results depending on context window, model version, or prompt variation.

Replayability: The V8 event-sourced model means every action can be replayed and every decision reconstructed. Prompt-based orchestration cannot be replayed deterministically.

Governance: The agent contract model defines explicit allowed/blocked actions, mandatory validators, and autonomy lane policies. A Super Agent orchestrating in prompt space could bypass these controls.

12.3 The Practical Difference

When a user says "Draft an appeal for the UHC denial on Elena Richardson's claim":

EVA as Super Agent (risky): EVA decides to retrieve the denial details, find relevant notes, look up the payer policy, draft the appeal text, and present it — all orchestrated by EVA's reasoning in the prompt.

EVA as conversational interface (correct): EVA recognizes the intent (appeal draft), identifies the subject (Elena Richardson's UHC denial), and invokes the denial appeal workflow. The workflow engine triggers the appeal-draft workflow step, which requests a context pack from the Context Assembly Service. The appeal-draft agent receives the governed context pack, produces a draft with a decision trace, and returns it as a Yellow-lane recommendation. EVA presents it conversationally while the right panel shows the full appeal draft with edit controls.

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Same user experience. Completely different governance and reliability model.

12.4 Mental Model

EVA is the face of the platform, not the brain. The brain is the workflow engine, context assembly, agent runtime, and validator framework. EVA is how humans interact with that brain through natural language.

13. Engineering Components

Component	Responsibility
EVA Session Service	Conversation state, focus entity tracking, right panel state, notification queue. Redis-backed stateful session per user.
Intent Resolver	LLM-powered intent parsing with entity extraction, resolution against platform state (patient index, DGS), confidence scoring, capability routing.
Capability Registry	Configuration artifact mapping resolved intents to workflow APIs, DGS endpoints, analytics pipeline, read-model queries. New capabilities added by registration.
Response Generator	LLM converts structured platform results into conversational responses with action chips, inline data cards, and panel navigation hints.
Right Panel Coordinator	WebSocket-based navigation commands to the right panel MFE. Sends screen, entityId, viewType, and highlight hints.
Notification Router	Event subscription, user routing, tier classification (Ambient/Nudge/Interrupt), and session push via WebSocket.

14. Intent Resolution Pipeline

14.1 Four-Stage Pipeline

Stage 1 — Classify intent: Data query, command, analytics query, workflow request, help request, or conversational filler. Uses a smaller/faster LLM (e.g., Claude Haiku) with a structured JSON prompt.

Stage 2 — Extract entities: Patient name, episode, claim, denial, payer, time period, metric. From "John Doe is here" → patient = John Doe, intent = check-in.

Stage 3 — Resolve against platform state: "John Doe" might match multiple patients. The resolver checks the tenant's patient index. If ambiguous, EVA asks a clarifying question. If unambiguous, proceeds.

Stage 4 — Map to capability: The Capability Registry routes the resolved intent to the appropriate platform API.

14.2 Key Design Point

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The LLM is used for natural language understanding (parsing intent from messy human language), not for deciding what to do. The Capability Registry decides what to do. The LLM tells the registry what the user meant. If confidence is low, EVA asks a clarifying question rather than guessing.

14.3 Capability Registry Examples

User Intent Pattern	Platform Capability	Type
"Check in [patient]"	Check-in workflow API	Workflow
"What's [patient]'s [attribute]?"	Domain Graph Service query	Data query
"Show me [metric] by [dimension]"	Conversational analytics (C.15)	Analytics
"Draft an appeal for [denial]"	Denial appeal workflow	Workflow
"What are my tasks today?"	Work item read model	Read model

15. Session State, Execution, and Response Generation

15.1 EVA Session State

The EVA Session Service is the stateful backbone of every user interaction. It maintains a synchronized view of what the user is discussing with EVA, what they are viewing in the right panel, and what entity they are working with — ensuring every EVA response is contextual regardless of whether the activity was initiated through EVA or the right panel directly.

The session maintains: user identity/role/tenant, interaction mode (hybrid/collapsed-AI/collapsed-context), notification queue, active workflow references, and a Context Stack where each entry contains the conversation context (conversation history as ordered turns), focus entity (type, ID, resolved timestamp), and panel context (screen, entityId, hasUnsavedWork) for that interaction. The session lives in Redis, not in the LLM context window. The LLM is stateless, invoked fresh per turn with the Active context entry's three layers injected.

15.1.1 Session Service with Context Stack

The Session Service manages the **Context Stack** — the primary data structure that tracks every meaningful interaction across both panels during a user's session.

Context Stack

Each entry in the stack carries its own unique `context_id` and each context stack entry must contain the following three context layers:

Conversation context — What EVA has been discussing with the user. Persisted across the conversation thread within this context entry.

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Panel context — What the user is currently viewing in both panels. Updated in real-time via the bidirectional WebSocket as the user navigates or takes actions in the structured UI.

Focus entity — The specific entity the user is working with. Updated by both conversation events and panel events.

The Intent Resolver consumes all three layers from the Active context entry and uses a priority hierarchy to disambiguate when the layers diverge: **ambiguous questions follow the panel context** (the user navigated deliberately), **explicit entity references follow the entity** regardless of panel state.

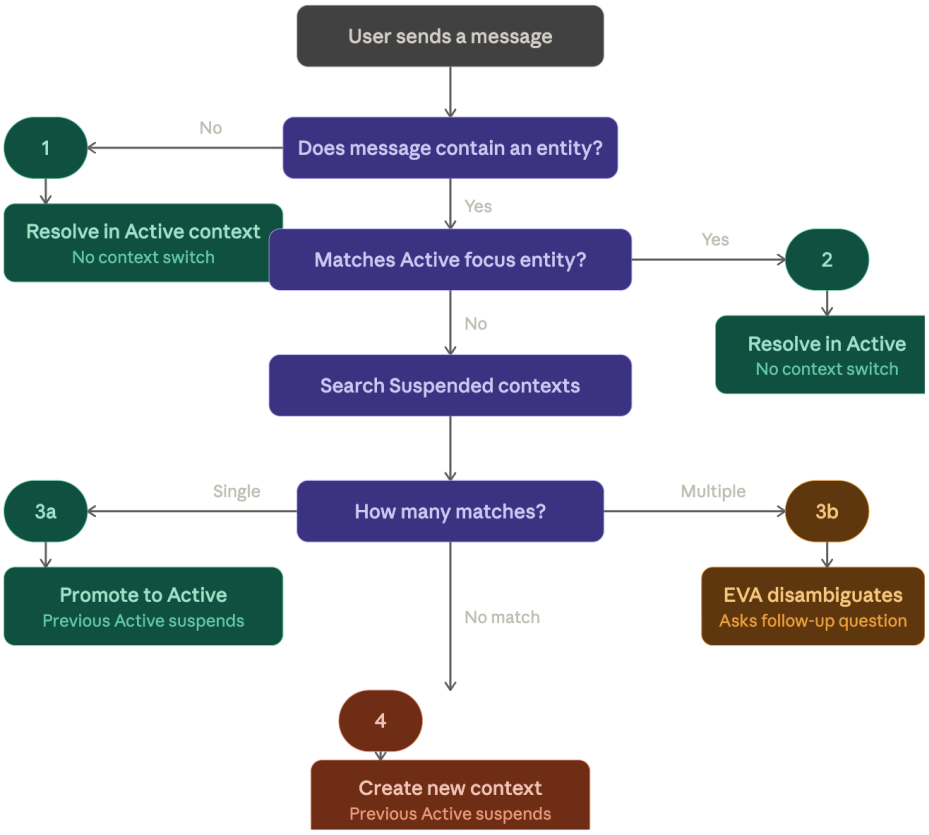
Context Entry in the Stack has three Lifecycle States:

Active — Current working context. Exactly one at any time. The Intent Resolver reads the three context layers from this entry to resolve every user message.

Suspended — Previous contexts, reactivatable on entity match. Retain a full three-layer state so the user can resume with both panels restored to the correct state.

Archived — No longer eligible for reactivation. Immutable. Exist solely for audit and history.

Entity Resolution:



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When the user sends a message, the Intent Resolver first resolves the entity against the Context Stack. If the stack resolves the entity, the Intent Resolution Pipeline (Section 14) proceeds from Stage 3 onward using the stack-resolved entity. If the stack does not resolve the entity (no match in Active or Suspended), Stage 3 of the pipeline resolves against the platform state (patient index, DGS) as specified in Section 14.1.

1. **No entity in the message.** If the user's message contains no entity (e.g., "what are my denial rates this month?"), it resolves in the current Active context. No context switch.
2. **Check Active context.** If the entity matches the focus entity in the Active entry, resolve there.
3. **Search Suspended contexts.** If no match in Active:
    - **Single match** → Promote to Active. Previously Active moves to Suspended.
    - **Multiple matches** → EVA asks a disambiguating follow-up.
4. **Create new context.** If no match is found, push a new Active entry. Previously Active moves to Suspended.

Scoping and Persistence:

The Context Stack is scoped to a single browser tab and persisted server-side against a browser session ID for crash recovery. Suspended contexts beyond a configurable threshold transition to Archived. This threshold needs to be determined.

15.1.2 Panel Synchronization

The Session Service keeps the panel context layer within the Active context entry synchronized with actual panel state through a **bidirectional WebSocket** via the Panel Coordinator.

The Session Service is reactive — it consumes these events as they arrive and updates the panel context layer in the Active context entry. The Intent Resolver, Response Generator, and Quick Action Generator all read from the Active entry, so every EVA output reflects the user's current state across all three layers.

EVA Behavior on Panel-Initiated Changes:

When the user navigates or takes an action in the right panel independently, three rules govern EVA's response:

1. **EVA does NOT proactively announce the change.** She does not say "I see you've moved to Today's Schedule!" That would be annoying — the user knows where they navigated. It would feel like the system is narrating their actions.
2. **EVA does silently update the panel context layer and focus entity in the Active context entry.** If the user's next message is a question, EVA uses the current panel context, not the stale conversation context. If the user asks "who's my next patient?" EVA answers from the schedule context, not from the John Doe referral context. The Intent Resolver (Section 14) receives both the user's message and the current session state. The session state tells the resolver: the user is looking at the daily schedule. So "next patient" resolves against the schedule, not against the referral.

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3. **EVA does preserve the conversation history.** The prior discussion is still in the conversation thread. If the user scrolls up or says "go back to John Doe's referral," EVA has the full context to resume. The Panel Coordinator sends a navigation command to return the right panel to the previous screen.

Unsaved Work Protection:

When the Session Service receives a `panel.unsavedWorkChanged` event with `hasUnsavedWork: true`, it records this in the Active context entry's panel context. If EVA subsequently needs to navigate the right panel (due to a conversation topic change or entity switch), EVA must prompt the user before proceeding: "You have unsaved changes in the referral form. Would you like to save before I switch to the analytics view?" This implements the unsaved work protection rule defined in through the Context Stack model.

Scenario 1 — User Navigates via the Green Bar:

User is in hybrid mode. The Active context has focus entity "John Doe referral," conversation about the referral intake, and panel showing the referral management screen. The user clicks "Today's Schedule."

Right Panel MFE → Panel Coordinator (WebSocket, bidirectional) → EVA Session Service receives:

{ eventType: "panel.navigated", source: "user", newScreen: "dailySchedule", newFocusEntity: null, previousScreen: "referralManagement", previousFocusEntity: { type: "referral", id: "ref-12345" }, hasUnsavedWork: false }

The EVA Session Service updates the Active context entry:

Active Context Entry BEFORE: conversationContext: "referral-intake", panelContext: { screen: "referralManagement", entityId: "ref-12345" }, focusEntity: { type: "referral", id: "ref-12345", label: "John Doe referral" }

Active Context Entry AFTER: conversationContext: "referral-intake" (unchanged), panelContext: { screen: "dailySchedule", entityId: null }, focusEntity: null (schedule is not entity-specific)

The three behavior rules above apply.

Scenario 2 — User Takes a State-Changing Action:

User checks in Maria Lopez directly via the schedule view, bypassing EVA. The state change flows through the normal architecture (MFE → BFF → canonical event → projections). Separately, the MFE notifies the Session Service:

Right Panel MFE → Panel Coordinator (WebSocket) → EVA Session Service receives:

{ eventType: "panel.actionPerformed", source: "user", actionType: "patientCheckIn", entity: { type: "appointment", id: "apt-maria-9am" }, screen: "dailySchedule" }

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The Session Service updates the Active context entry's panel context to reflect the check-in. The conversation context is unchanged — no conversation turns were exchanged. The focus entity may shift to Maria depending on whether the action is treated as an entity selection. Because the panel context now reflects Maria's checked-in state, EVA won't re-run the check-in workflow if the user mentions Maria — she'll acknowledge it's done and offer the next step. Quick actions update — "Check in Maria Lopez" is replaced by "Collect Maria's copay." An optional Tier 1 ambient acknowledgment may appear per notification tier rules.

15.1.3 Context Divergence Resolution

Context divergence occurs when the panel context and conversation context within the Active entry are about different things. The Intent Resolver applies the priority hierarchy from 15.1.1.

15.2 Capability Execution

Once EVA has a resolved intent, it invokes the platform capability and streams results as they arrive. For workflow invocations, EVA calls the workflow API and the workflow engine (Temporal) manages all steps, agents, validators, and state. For data queries, EVA calls the Domain Graph Service through the BFF. For analytics, EVA invokes the C.15 pipeline. EVA does not orchestrate multi-step workflows — she invokes them and presents results.

15.3 Response Generation

The Response Generator converts structured results into conversational text plus: 2–4 suggested action chips, a right panel navigation hint, and inline data cards where appropriate. The right panel MFE receives navigation commands via WebSocket and fetches its own data independently.

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16. Learning Loops

Layer	What Improves	Data Source	Ships	Mechanism
1. User Preferences	Quick actions, response style, morning greeting, task prioritization for individual users	Behavioral observation: what the user asks first, which actions they click, which chips they use	Phase 1–2	Structured user profile injected into intent resolver + response generator
2. Practice Patterns	Practice-level proactive suggestions, denial playbook emphasis, workflow defaults	Analytical surface aggregation: top denial codes, payer patterns, no-show rates, common workflows per practice	Phase 2–3	Practice intelligence store read at session init
3. Outcome-Linked	Appeal strategy effectiveness, denial prevention, recommendation quality	Decision traces: what was recommended, accepted/overridden, downstream outcome	Optimization	Outcome correlation analysis → playbook updates (human-reviewed)
4. Intent Tuning	Intent resolution accuracy, entity recognition, practice-specific vocabulary	Correction logs: when users rephrase after EVA misunderstands	Ongoing	Periodic review → resolver prompt/logic tuning
5. Predictive	No-show prediction, denial risk scoring, care dropout prediction	Full analytical surface historical data	6–12 mo post-launch	ML models trained on analytical surface, surfaced by EVA proactively

16.1 Making Learning Visible

- EVA acknowledges what she's learned: "I've noticed you usually start with referrals on Monday mornings. Want me to pull those up first?"
- Monthly practice intelligence summaries: "Here's what I learned about your practice this month."
- The "why" behind suggestions: "I'm suggesting this appeal approach because it's succeeded in 72% of similar cases."
- Quick actions that evolve: by week four, actions reflect the user's actual workflow, not system defaults.

16.2 Learning Guardrails

Learning must not autonomously change behavior without human review.

One user's preferences must not be applied to another user without governance.

Learned patterns must be presented as recommendations with evidence, not as rules.

User/practice profiles must not contain PHI. They store behavioral and operational patterns.

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17. Concrete Walkthrough: "John Doe Is Here"

1. Session Service receives the message and adds it to conversation history.
2. Intent Resolver calls the LLM with message + session context. Returns: {intentCategory: "start_workflow", subIntent: "patient_check_in", entities: [{type: "patient", value: "John Doe"}], confidence: 0.95}.
3. Intent Resolver resolves "John Doe" against the tenant's patient index. If ambiguous, asks for clarification.
4. Capability Registry maps the intent to the check-in workflow API.
5. EVA invokes the check-in workflow. The workflow engine runs: verify identity, invoke eligibility agent with context pack, check outstanding balance, update schedule, notify provider. Each step emits canonical events.
6. Results stream back. Response Generator produces: "John Doe is checked in. Eligibility verified with Blue Shield — $25 copay, 22 sessions remaining. Dr. Aris has been notified." Chips: "Collect copay" / "View patient profile" / "Any notes on the referral?"
7. Right Panel Coordinator sends navigation command: {screen: "patientDashboard", entityId: "pat_john_doe", viewType: "checkInSummary"}.
8. Session Service updates focus entity to pat_john_doe and right panel state to patientDashboard.

Behind the scenes: a workflow ran, three agents executed, five events were emitted, and decision traces were recorded. EVA orchestrated none of it. The workflow engine did. EVA made it feel seamless.

18. Delivery Phasing

Phase	EVA + UX Deliverables	Dependencies
Phase 0	No EVA deliverables. Platform scaffold must be operational first.	Event backbone, graph, DGS, workflow engine, CAS skeleton.
Phase 1	Three-mode shell with panel coordination. EVA Session Service. Intent Resolver (initial). Capability Registry (referral intake, patient lookup). Response Generator. Right Panel Coordinator. Basic notification (Tier 1). Referral intake workflow screen.	Phase 0 platform, referral intake workflow, basic DGS endpoints.
Phase 2	Expanded capabilities (scheduling, eligibility, authorization). Full notification tiers. Layer 1 learning. Quick action personalization. Provider session with ambient scribe UX. Owner analytics dashboard integration.	Phase 1 expansion (scheduling, eligibility workflows). Ambient scribe pipeline.
Phase 3+	Full conversational analytics through EVA (C.15). Layer 2 practice patterns. Expanded action chip generation. Front office calendar integration. Intent resolver tuning.	Analytical Data Surface, semantic layer, charge/billing workflows.
Optimization	Layer 3 outcome-linked learning. Layer 5 predictive models (no-show, denial risk, dropout). Monthly practice intelligence summaries.	Sufficient historical data in analytical surface and decision traces.

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19. Architectural Invariants

EVA-1: EVA must not orchestrate multi-step workflows in prompt space. All multi-step processes run in the durable workflow engine.

EVA-2: EVA must not bypass the governed agent runtime. Every agent invocation uses a context pack with validators and lane policies.

EVA-3: EVA must not mutate business state directly. All state changes flow through workflow → command → event.

EVA-4: EVA's failure mode must be safe. A mis-parse results in a clarifying question never an ungoverned action.

EVA-5: Every EVA interaction must be traceable end-to-end through the platform's observability model.

EVA-6: Notification tiers must be system-governed, not user-configured.

EVA-7: Quick actions and action chips must be generated from context packs and task profiles, not hardcoded.

EVA-8: Proactive panel expansion must not disrupt clinical work. Use notification badges, not forced layout changes.

EVA-9: Learning loop updates must go through human review before deployment.

EVA-10: EVA is the conversational face of the platform. The brain is the workflow engine, context assembly, agent runtime, and validator framework.

20. Handbook Cross-References

Document Topic	Handbook Section
BFF and API gateway baseline	Appendix C, Section C.7
Microfrontend experience baseline	Appendix C, Section C.6
Context Assembly Service (12-step waterfall)	Section 8 (8.1–8.19, 8.6A, 8.8.1–8.8.4)
Agent contract template and delegation	Section 9.12A–12C
Agentic execution patterns (1–3)	Section 9.12B
Autonomy lanes (Green/Yellow/Red)	Sections 9.5–9.9
Decision trace model	Section 9.13
Human override model	Section 9.14
Streaming Assist / Ambient scribe	Section 9.10.2, 9.10.3
Conversational analytics	Appendix C, Section C.15
Workflow orchestration (Temporal)	Section 10, Section 10.2A
Human-in-the-loop design	Section 10.12
Analytical Data Surface	Appendix B, Analytical Surface Addendum
Domain Graph Service	Section 7 (7.1–7.21)
Observability model	Section 13

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